HARBOR HARNESS / TERMINAL-BENCH 2.1 TASK-PROVIDED VERIFIERS

68 tasks. Two runs. 31% lower median cost.

Exploratory pooling of two native Claude Code and PennyRouter runs, using the **Harbor harness and Terminal-Bench task-provided verifiers**. 272 trials; 63 tasks contribute to paired statistics.

31.1%

LOWER MEDIAN MODEL-CALL COST PER TASK

95% interval for median change
−35.2% to −25.5%

Primary endpoint / 63 pooled tasks
Wilcoxon $p = 5.05 \times 10^{-8}$

51 / 63	12.1%	63 tasks
Tasks with lower model-call cost	Lower spending across 116 eligible repetitions	53 with two eligible reps; 10 with one

Model-call cost, summed



$1.170 less across 116 eligible task-repetitions per arm. Actual summed spend: -12.1%. Notional model-call USD, not a subscription bill.

Quality at a glance



Penny scored higher on 11 tasks, lower on 6, and tied on 46. Mean task-level reward was 0.881 through Penny versus 0.841 natively.

Both source runs passed their full evidence audits. This is a post-hoc pooled analysis, not a preregistered k=2 study. Original rewards and source files are unchanged.

Where the difference appears.

Median task-level cost fell 31.1%. The sum of per-task medians fell 10.4%; actual spending across eligible repetitions fell 12.1%. These use different task weights.

ENDPOINT	MEDIAN CHANGE	95% CI, MEDIAN %	SUM OF TASK MEDIANS	P-VALUE
Model-call cost	−31.1%	[−35.2, −25.5]	−10.4%	5.05e−08
Total tokens	−20.9%	[−23.0, −9.8]	+7.7%	8.99e−04
Output tokens	+0.9%	[−2.3, +13.7]	+25.1%	0.056
Fresh input tokens	+16.7%	[+0.0, +25.0]	−81.4%	1.25e−04
Cache-read tokens	−19.6%	[−21.6, −7.8]	+8.8%	0.002
Cache-write tokens	−24.4%	[−34.4, −14.4]	−7.2%	1.83e−06
Model calls	+0.0%	[+0.0, +16.7]	+15.9%	0.050
Agent execution time	+12.9%	[+4.9, +21.5]	−25.3%	0.011

7.50M → 8.08M

Sums of per-task median token counts, native to Penny. Cache-read tokens: 6.60M → 7.18M. These are not totals across all attempts.

203,403.5 → 254,503.0

Sums of per-task median output tokens, native to Penny: +25.1%. The median task-level change was +0.9%.

How to read the statistics

Within each task, take each arm's median across eligible repetitions, then compare Penny / native - 1. A repetition is eligible only when both arms completed. Tasks are the statistical unit; 272 trials are not treated as independent tasks.

The median and 95% bootstrap interval describe task-level changes. P-values use paired Wilcoxon signed-rank on per-task log ratios; the cost Hodges-Lehmann estimate is -27.9%. Table sums add 63 per-task medians; the first-page cost bars sum spending over 116 eligible repetitions.

Agent execution time uses Harbor's agent-execution span, excluding container build and installation. With both eligible repetitions required (53 tasks), median cost fell 30.2% and summed task cost fell 14.2%. Quality tests do not establish equivalence or non-inferiority.

Methodology, quality & audit.

Harbor execution. Terminal-Bench verification.

Two independent k=1 runs cover the same 68 tasks and arms. Their 272 finished trials are pooled here post-hoc. The full evidence audit passed in each source run. Original task rewards, graders, and source manifests are unchanged; this post-hoc analysis does not create a new preregistered experiment.

TASK-LEVEL MEAN REWARDS / 63 POOLED TASKS

0.841	0.881
Native mean reward	Penny mean reward



11	46	6
Penny higher	Equal reward	Penny lower

Sign test p = 0.332. No significant difference detected. Values are mean task-level rewards after pooling, not all-attempt pass rates. No non-inferiority margin was declared.

EVALUATION SCOPE

68 = 53 + 10 + 5

53 tasks have two eligible repetitions; 10 have one; 5 have neither. 116 task-repetitions are eligible, 20 excluded.

All-attempt outcomes: Native passed 103/136 attempts (75.7%); Penny passed 104/136 (76.5%). Errors remain scored failures. All 272 rewards are present.

Pooling scope: 63 tasks contribute to paired statistics. Only eligible repetitions contribute to their per-task medians. Both-repetitions-only results are saved separately as a sensitivity check.

11 / 11 audit checks accepted

Both source runs passed all 11 evidence checks. These are source-run audits, not a new pooled audit or preregistration. No adoption marker was declared.

Completeness PASS Two full source runs: all 272 planned trials finished.	**Wire capture** PASS Decodable model calls for every completed included trial.	**Routing** PASS Captured calls match their declared arm host.
Version pins PASS Harness version and model match arm specifications.	**Adoption marker** N/A* Not checked: no adoption marker was declared.	**Task skills** PASS No task-local skills were installed in either arm.
Cache isolation PASS Each trial has its own reference line in requests.	**Account plan** PASS Paired arms use the same recorded Pro account plan.	**Environment** PASS Included completed trials have matching paired package inventories.
Host isolation PASS No host home-directory path appears in requests.	**Pairing** PASS Finished pairs have both arms; excluded tasks did not run in this manifest.	**Interpretation** POOLED ANALYSIS *Adoption is a no-op check, accepted by the auditor without testing a marker.

Setup: Terminal-Bench 2.1 / Sonnet 5.5, medium / Claude Code 2.1.302 / no task skills / two independent runs of 68 tasks / Pro plan / rates pinned 2026-10-15. Source seeds: 20261019 and 20261020. Selection: 68 of 89 tasks after 17 resource exclusions and 4 reference failures. The first run was registered; the second was not. This pooling was selected post-hoc. Recommended reproduction route: hosted https://api.pennyrouter.com with your own linked account.

Every task. Every pooled reward.

Tasks 01-20 of 68, in first-run order. Rewards and costs are per-arm medians over eligible repetitions. N/A means no eligible repetition, not a missing original reward.

#	TASK	NATIVE	PENNY	COST Δ	SCOPE
01	modernize-scientific-stack	1.000	1.000	+0.4%	2 eligible reps
02	overfull-hbox	0.500	1.000	−30.2%	2 eligible reps
03	torch-tensor-parallelism	1.000	1.000	−40.7%	2 eligible reps
04	git-multibranch	1.000	1.000	−11.3%	2 eligible reps
05	build-pmars	1.000	0.500	−32.7%	2 eligible reps
06	largest-eigenval	1.000	1.000	+39.0%	2 eligible reps
07	sparql-university	1.000	1.000	−30.1%	2 eligible reps
08	make-doom-for-mips	N/A	N/A	N/A	No eligible reps
09	db-wal-recovery	1.000	0.000	−22.7%	2 eligible reps
10	nginx-request-logging	1.000	1.000	−37.8%	2 eligible reps
11	build-cython-ext	1.000	1.000	+2.6%	2 eligible reps
12	openssl-selfsigned-cert	1.000	1.000	−40.0%	2 eligible reps
13	hf-model-inference	1.000	1.000	−35.2%	2 eligible reps
14	polyglot-c-py	1.000	1.000	−37.3%	2 eligible reps
15	kv-store-grpc	1.000	1.000	−36.3%	2 eligible reps
16	model-extraction-relu-logits	1.000	0.500	+4.1%	2 eligible reps
17	pytorch-model-recovery	0.500	1.000	−31.7%	2 eligible reps
18	regex-log	1.000	1.000	−29.9%	2 eligible reps
19	multi-source-data-merger	1.000	1.000	−29.0%	2 eligible reps
20	vulnerable-secret	N/A	N/A	N/A	No eligible reps

Each eligible repetition has both arms marked complete. Per-task values use the median across the one or two eligible repetitions. Tasks with no eligible repetitions are excluded from paired statistics; all 272 original attempt outcomes are retained in supporting data.

Every task. Every pooled reward.

Tasks 21-40 of 68, in first-run order. Rewards and costs are per-arm medians over eligible repetitions. N/A means no eligible repetition, not a missing original reward.

#	TASK	NATIVE	PENNY	COST Δ	SCOPE
21	count-dataset-tokens	1.000	1.000	−33.2%	1 eligible reps
22	raman-fitting	0.000	1.000	−11.5%	2 eligible reps
23	pytorch-model-cli	1.000	1.000	−25.5%	1 eligible reps
24	sqlite-with-gcov	1.000	1.000	−40.5%	1 eligible reps
25	fix-git	1.000	1.000	−20.7%	2 eligible reps
26	cobol-modernization	0.500	1.000	−26.0%	2 eligible reps
27	merge-diff-arc-agi-task	0.500	1.000	−5.1%	2 eligible reps
28	polyglot-rust-c	1.000	1.000	−36.2%	2 eligible reps
29	torch-pipeline-parallelism	1.000	1.000	−51.2%	1 eligible reps
30	git-leak-recovery	1.000	1.000	−35.0%	2 eligible reps
31	pypi-server	1.000	1.000	−37.2%	2 eligible reps
32	extract-elf	1.000	1.000	−41.4%	2 eligible reps
33	prove-plus-comm	1.000	1.000	−45.9%	2 eligible reps
34	gcode-to-text	1.000	1.000	+61.7%	1 eligible reps
35	chess-best-move	0.500	0.000	−47.3%	2 eligible reps
36	cancel-async-tasks	1.000	1.000	−36.6%	2 eligible reps
37	sqlite-db-truncate	1.000	1.000	−71.7%	2 eligible reps
38	write-compressor	1.000	1.000	−23.2%	2 eligible reps
39	headless-terminal	1.000	1.000	−37.7%	2 eligible reps
40	log-summary-date-ranges	1.000	1.000	−33.0%	2 eligible reps

Each eligible repetition has both arms marked complete. Per-task values use the median across the one or two eligible repetitions. Tasks with no eligible repetitions are excluded from paired statistics; all 272 original attempt outcomes are retained in supporting data.

Every task. Every pooled reward.

Tasks 41-60 of 68, in first-run order. Rewards and costs are per-arm medians over eligible repetitions. N/A means no eligible repetition, not a missing original reward.

#	TASK	NATIVE	PENNY	COST Δ	SCOPE
41	password-recovery	1.000	1.000	−19.1%	2 eligible reps
42	adaptive-rejection-sampler	1.000	0.500	+15.4%	2 eligible reps
43	configure-git-webserver	0.500	1.000	−26.8%	2 eligible reps
44	fix-code-vulnerability	1.000	1.000	−18.6%	2 eligible reps
45	sanitize-git-repo	0.500	1.000	−36.1%	2 eligible reps
46	gpt2-codegolf	1.000	1.000	+36.3%	2 eligible reps
47	query-optimize	1.000	0.500	−19.2%	2 eligible reps
48	code-from-image	1.000	1.000	−36.6%	2 eligible reps
49	large-scale-text-editing	1.000	1.000	−43.9%	2 eligible reps
50	constraints-scheduling	1.000	1.000	−37.7%	2 eligible reps
51	financial-document-processor	1.000	1.000	−39.4%	1 eligible reps
52	break-filter-js-from-html	N/A	N/A	N/A	No eligible reps
53	crack-7z-hash	N/A	N/A	N/A	No eligible reps
54	path-tracing-reverse	1.000	1.000	+10.4%	2 eligible reps
55	llm-inference-batching-scheduler	1.000	1.000	−33.9%	1 eligible reps
56	rstan-to-pystan	1.000	1.000	−48.4%	1 eligible reps
57	custom-memory-heap-crash	0.500	1.000	+7.0%	2 eligible reps
58	feal-linear-cryptanalysis	1.000	1.000	+10.1%	1 eligible reps
59	protein-assembly	0.500	1.000	−31.8%	2 eligible reps
60	path-tracing	1.000	1.000	+181.8%	1 eligible reps

Each eligible repetition has both arms marked complete. Per-task values use the median across the one or two eligible repetitions. Tasks with no eligible repetitions are excluded from paired statistics; all 272 original attempt outcomes are retained in supporting data.

Every task. Every pooled reward.

Tasks 61-68 of 68, in first-run order. Rewards and costs are per-arm medians over eligible repetitions. N/A means no eligible repetition, not a missing original reward.

#	TASK	NATIVE	PENNY	COST Δ	SCOPE
61	dna-assembly	0.500	1.000	−17.6%	2 eligible reps
62	filter-js-from-html	0.000	0.000	−37.9%	2 eligible reps
63	mailman	1.000	1.000	−27.0%	2 eligible reps
64	feal-differential-cryptanalysis	1.000	1.000	+15.4%	2 eligible reps
65	extract-moves-from-video	N/A	N/A	N/A	No eligible reps
66	make-mips-interpreter	0.000	0.500	−0.9%	2 eligible reps
67	dna-insert	0.000	0.000	−31.1%	2 eligible reps
68	mteb-retrieve	0.000	0.000	−35.7%	2 eligible reps

Source record / pooled analysis

Source experiment IDs: proxy-tb21-bounded; proxy-tb21-repeat-20261020

Supporting files: REPORT.md, stats.json, outcomes.json, metrics.json, trials.json, audit.json, run.json, sources.json, both-repetitions-only.json, source-runs/

Published data: benchmarks/terminal-bench/results/pennyrouter-tb21-20261019-20261020/. Original trials, rewards and source manifests are unchanged.